

8 (short)

(a) Explain the functions of STATUS Register, Working Register (W) and TRISIO Register in PIC12F675 microcontroller. [3]

(b) Before the following PIC assembly code is executed, the contents of memory locations 0x30 and 0x31 are the decimal numbers 20 and 50, respectively.

```
main
    movlw 0x31;
    movwf FSR;
    call sr;
    decf FSR, f;
    call sr;
end    sleep;
...
sr    rrf INDF, f;
    movlw 0x10;
    addwf INDF, f;
    return;
```

(i) What are the contents of Working Register (W) and the memory locations of 0x30 and 0x31 after the code has executed? [4]

(ii) Determine the time taken to execute the code assuming a 20 MHz operational clock. [3]